Explainable and Responsible Artificial Intelligence

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AI Governance Maturity

Maturity Model of **AI Governance**

Level 0:
No AI lifecycle governance

Level 1:
AI policies available to **guide** AI lifecycle governance

Level 2:
Common set of **metrics** to govern AI lifecycle

Level 3:
Enterprise **data** and **AI catalog**

Level 4:
Automated validation and **monitoring**

Level 5:
Fully **automated** AI lifecycle governance

https://www.ibm.com/blogs/watson/2021/06/ai-governance-maturity/

Level 0: No AI lifecycle governance

A At this level, each AI development team uses their own tools, and there are no documented central policies for AI development or deployment.

B This approach provides a lot of flexibility and is typical for organizations starting out on their journey to AI.

C However, it has the potential to introduce significant risks to the business if these models were to be deployed to production.

D Specifically, since there is no framework, it would be impossible to even evaluate the source of risk.

E Companies at this level tend to find scaling AI practices difficult.

F Hiring 10x data scientists does not lead to 10x increase in AI productivity due to inconsistencies.

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Level 1: AI policies available to guide AI lifecycle governance

1

A This level sees AI policies being created at either a line-of-business or enterprise level (such as a CDO or CRO) for constructing and deploying AI, as well as a common definition of information required before validating a model.

B However, as there is no enforcement of these policies, individual AI systems are still siloed with little consistency.

C In this stage, there is potential for misunderstanding the policies and in rare cases even subverting them, since there is no common monitoring framework to provide enforcement.

D The potential for risk here is also high.

E Companies at this level do not see many improvements in productivity, but they start to develop strategies for measuring successful AI.

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Level 2: Common set of metrics to govern AI lifecycle

2

A This level builds on level 1 by defining a standard set of acceptable metrics and a monitoring tool to evaluate models.

B This not only brings consistency among the AI teams, but also enables metrics to be compared across different development lifecycles.

C A common monitoring framework is typically introduced to track these metrics, to enable everyone in the organization to interpret them in the same way.

D This reduces the level of risk and improves transparency of information needed to make policy decisions or troubleshoot reliability in case of issues.

E Companies at this level usually have a central model validation team upholding the policies laid out by the enterprise during their validation process, so they start to see some productivity gains.

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Level 3: Enterprise data and AI catalog

3

A This level leverages the metadata information from level 2 to ensure that all assets in a model's lifecycle are available in an enterprise catalog with data quality insights and data provenance.

B With a single data and AI catalog, the enterprise can now trace the full lineage of data, models, lifecycle metrics, code pipelines and more.

C This also lays the foundation for making connections between the numerous versions of models to enable a full audit in compliance situations.

D It also provides a single view to a CDO/CRO for a comprehensive AI risk assessment.

E Companies at this level are able to clearly articulate risks related to AI and have a comprehensive view of the success of their AI strategy.

https://www.ibm.com/blogs/watson/2021/06/ai-governance-maturity/, https://www.ibm.com/products/knowledge-catalog

Level 4: Automated validation and monitoring

4

A This level introduces automation into the process to automatically capture information from the AI lifecycle.

B This information significantly reduces the burden on the data scientist (and other lifecycle participants) to manually document their actions, measurements, and decisions.

C This information also enables model validation teams to make decisions on an AI model, as well as allowing them to leverage AI-based suggestions.

D With this capability, an enterprise can significantly reduce the operations effort in documenting data and model lifecycles.

E It also removes any risks from mistakes along the lifecycle in terms of metrics, metadata, or versions of data/model being left out.

F Companies at this level start to see an exponential increase in productivity as they're able to consistently and quickly put AI models into production.

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Level 5: Fully automated AI lifecycle governance

5

A This level uses the automation from the previous step to automatically enforce enterprise-wide policies on AI models.

B This framework now ensures that enterprise policies will be enforced consistently throughout all models' lifecycles.

C At this point, an organization's AI documentation is produced automatically with the right level of transparency through the organization for regulators and, more importantly, for customers.

D This enables the team to prioritize the riskiest areas for more manual intervention.

E Companies at this level can be extremely efficient in their AI strategy while maintaining confidence in their risk levels.

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